STAT 7900: Special Topic: Causal Analysis



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STAT 7900

Recap Slides

1 QUESTION

What would happen if we intervened?

→

2 COUNTERFACTUALS

Define Y(1), Y(0) and the causal estimand

→

3 GRAPHS

Use a DAG to decide what must be adjusted for

→

4 INTERVENTIONS

Use an SCM and do(·) to define intervention worlds

Identification comes before estimation.

Today: SCM and why randomization makes the identification step unusually transparent.

Recap Slides

DESCRIPTIVE

What is happening?

How does blood pressure differ between smokers and non-smokers?

PREDICTIVE

What will happen?

Can age, BMI, and smoking predict future blood pressure?

CAUSAL

What would happen if we intervened?

Would smoking cessation lower blood pressure?

The scientific question determines the analysis.

Recap Slides

OBSERVED ASSOCIATION

$$E[Y \mid T = 1] - E[Y \mid T = 0]$$

$\neq$ **in general**

CAUSAL EFFECT

$$E[Y(1)] - E[Y(0)]$$



A common cause C creates the backdoor path $T \leftarrow C \rightarrow Y$.

Correlation can be a mixture of causal and noncausal association.

Recap Slides

Fundamental problem: only one world is observed; the other is counterfactual.

WORLD 1

T = 1

Potential outcome: Y(1)

WORLD 0

T = 0

Potential outcome: Y(0)

Individual effect: $Y_i(1) - Y_i(0)$

ATE: $E[Y(1) - Y(0)]$

Causal inference learns about missing counterfactuals.

Recap Slides

ATE

$E[Y(1) - Y(0)]$

Average effect in the target population

ATT

$E[Y(1) - Y(0) \mid T = 1]$

Effect among those actually treated

ATC

$E[Y(1) - Y(0) \mid T = 0]$

Effect among those actually untreated

CATE(x)

$E[Y(1) - Y(0) \mid X = x]$

Effect in a covariate-defined subgroup

Same data + different estimand → different causal question.

Four Assumptions

CONSISTENCY

Observed Y equals the potential outcome under the treatment actually received.

EXCHANGEABILITY

After conditioning on the right X, treatment carries no extra information about Y(t).

POSITIVITY

Within relevant X strata, both treatment options have positive probability.

Failure: no overlap / unsupported counterfactual.

SUTVA / NO INTERFERENCE

Treatment is well defined and one unit's treatment does not alter another's outcome.

Failure: hidden versions or spillovers.

These assumptions are not fully testable from the observed data alone.

Three Building Blocks

CHAIN

$$X \rightarrow Z \rightarrow Y$$

Open by default.
Blocked when conditioned on Z.

FORK

$$X \leftarrow Z \rightarrow Y$$

Open by default.
Blocked when conditioned on Z.

COLLIDER

$$X \rightarrow Z \leftarrow Y$$

Blocked by default.
Open when conditioned on Z.

Conditioning behaves differently in the collider.

Recap

Variable adjusted for	Graph role	What adjustment does
Confounder C	$T \leftarrow C \rightarrow Y$	Removes bias
Mediator M	$T \rightarrow M \rightarrow Y$	Blocks part of total effect
Collider C	$T \rightarrow C \leftarrow Y$	Creates association / bias

Regression syntax cannot tell you which causal role a covariate has.

Recap

R with dagitty

```
library(dagitty)

g <- dagitty("dag {
  T [exposure]
  Y [outcome]

  Age -> BP;  Age -> T
  BP  -> DM;  BP  -> T;  BP -> Y
  DM  -> T;   DM  -> BP3
  T   -> BP3; T   -> Y
  BP3 -> Adh; BP3 -> Y
  Pref -> Adh; Adh -> Y
}")

adjustmentSets(g, type = "minimal",
               effect = "total")
```

Output: { BP, DM }

What R is checking

Minimal set = { BP, DM }

BP is required to block
$T \leftarrow BP \rightarrow Y$.

DM blocks the remaining path
$T \leftarrow DM \rightarrow BP3 \rightarrow Y$.

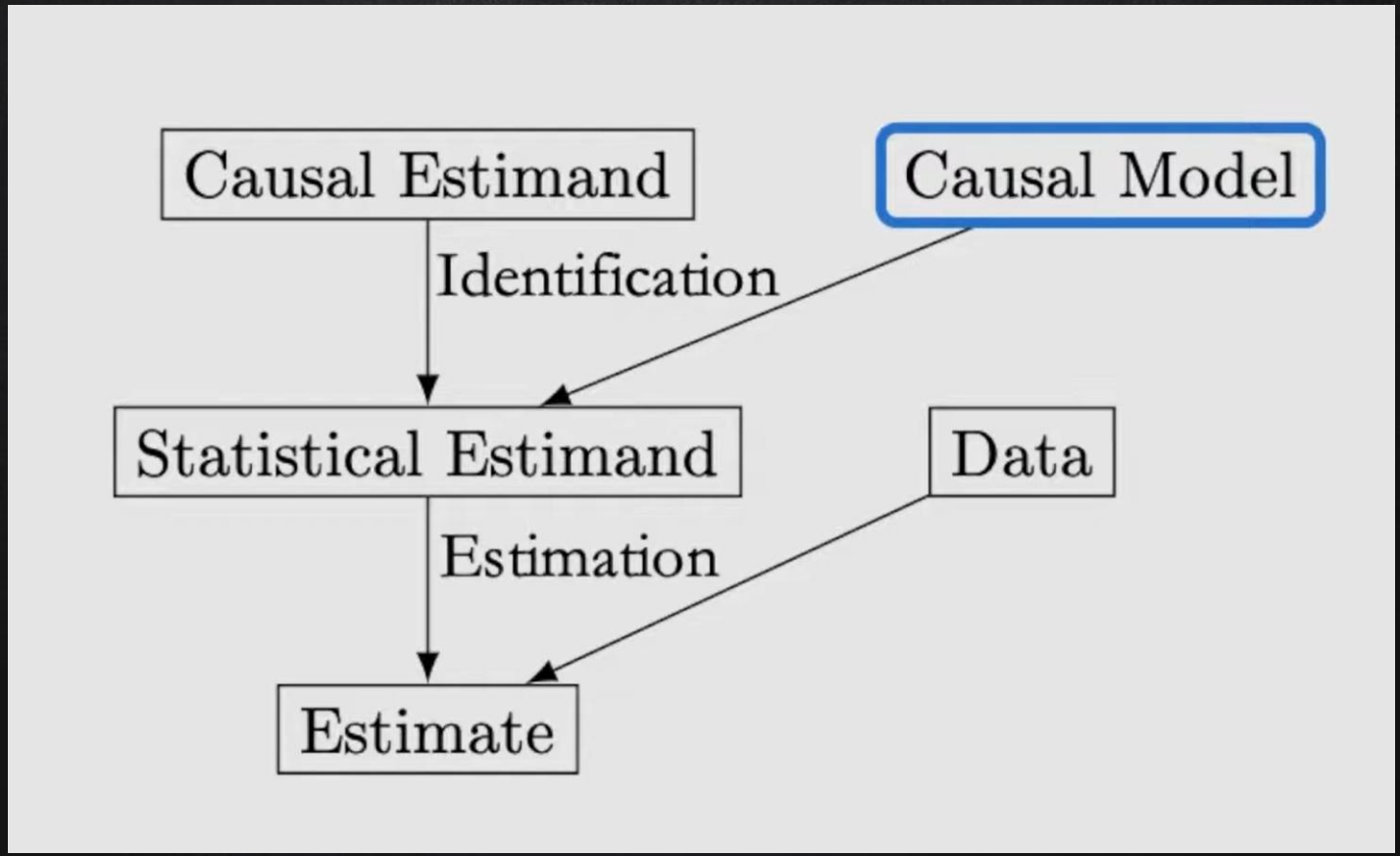
Age is not needed once BP is already in the set.

Do not adjust for BP3 or Adh: they are post-treatment.

STAT 7900: Quiz 3



Structural Causal Models (SCM)



Structural Causal Models (SCM)

From a causal picture to a model of how the world generates variables

DAG = causal map

SCM = causal map + generating rules

We will learn how to represent an intervention such as “force treatment = 1.”

Structural Causal Models (SCM)

A DAG already tells us:

- which variables directly cause which.
- which paths are causal or noncausal.
- what we may need to adjust for.

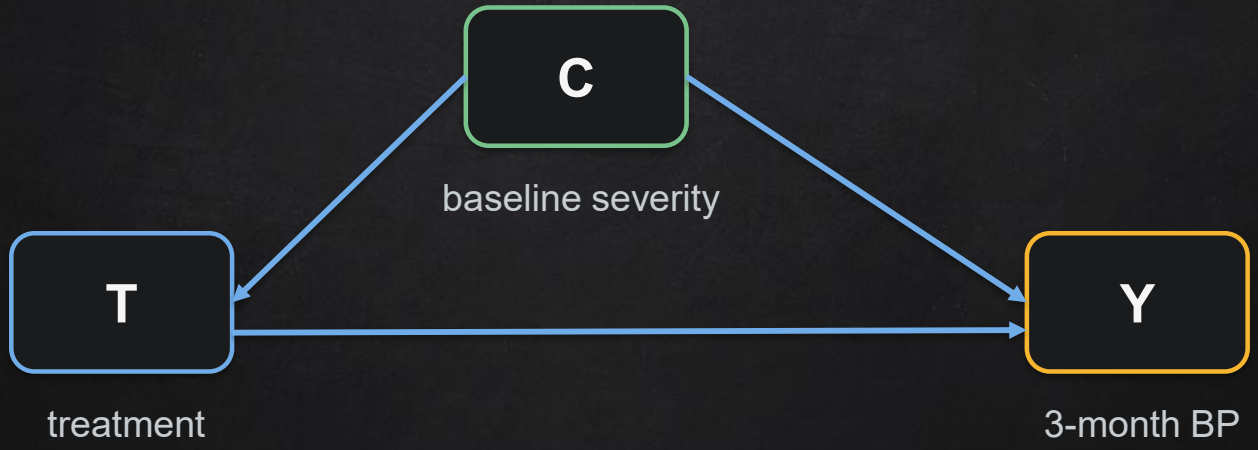
SCM adds something new:

- a rule for how each variable is generated.
- a precise meaning for “intervene”.
- a way to define the same person under different treatment choices.

SCM makes “what if we changed T?” mathematically precise.

Example

Clinical question: **Does an intensive blood-pressure treatment lower systolic BP after 3 months?**



Example

$C \rightarrow T$ Baseline severity can influence which treatment is chosen.

$C \rightarrow Y$ Baseline severity can also influence the outcome.

$T \rightarrow Y$ Treatment can causally change the outcome.

The DAG gives direction, but not yet the actual rule for generating T or Y.

Example

The graph does not tell us:

- How strongly does severity affect treatment choice?
- How strongly does treatment change blood pressure?
- What other background factors affect treatment or outcome?
- Exactly what should we change when we say “give treatment to everyone”?

SCM supplies those generating rules.

SCM

1
Generate baseline severity C

2
Generate treatment T using C + other factors

3
Generate outcome Y using T, C + other factors

Each recipe says: “given the direct causes of this variable, how is this variable produced?”

U Variables

U means background factors not explicitly represented as nodes in our simple DAG.

U_C	U_T	U_Y
genetics, long-term history, measurement noise	clinician preference, logistics, patient choice	other causes of the outcome

We use U because no model includes every cause in the universe.

Important: a U variable can be measured or unmeasured in reality. Here it simply means “outside the simplified system we drew.”

Treatment Assignment

$$T := f_T(C, U_T)$$

This does NOT mean treatment is random.

It says treatment is produced by some rule that can depend on baseline severity C and other treatment-related factors U_T.

f_T = Treatment Decision Rule

Outcome

$$Y = 120 + 10C - 6T + U_Y$$

One possible numerical rule:

- Every 1-unit increase in baseline severity raises Y by 10 units.
- Treatment lowers Y by 6 units, holding the other causes fixed.
- U_Y represents other causes of Y not shown explicitly.

What makes an SCM causal

A regression can describe association.

An SCM makes an additional scientific claim:

If we externally change one variable, the other generating rules remain the rules of the system.

That is what allows the model to answer intervention questions.

Key Operation: Intervention

Normal observational world:

$$T = f_T(C, U_T)$$

Treatment follows the usual decision process.

Intervention world:

$$T := 1$$

We replace the treatment rule and force treatment to 1.

Everything else in the SCM stays the same.

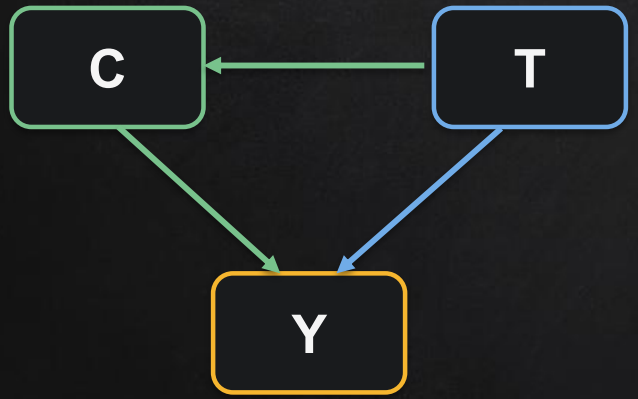
Do Calculus

$$do(T = 1)$$

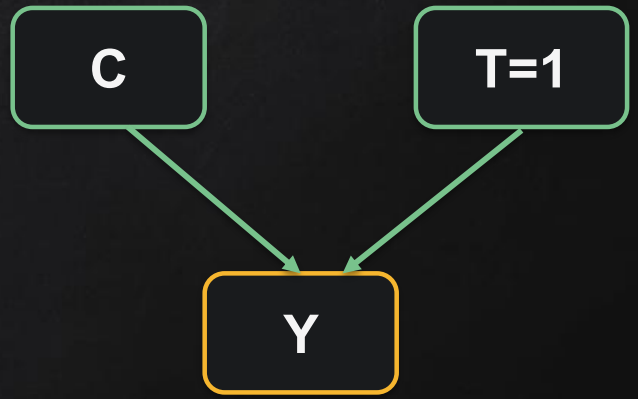
“Set treatment to 1 from outside the system.”

- Do not let baseline severity decide T.
- Do not let clinician preference decide T.
- Simply assign T = 1, then let the rest of the model run.

Before intervention



After do(T = 1)



Intervention vs Conditioning

Conditioning: T = 1
Look at the people who happened to receive treatment.

Their baseline characteristics may differ from untreated people.

Intervention: do(T = 1)
Force treatment to 1 for the target population.

The usual treatment-selection process is bypassed.

$P(Y \mid T = 1)$ and $P(Y \mid do(T = 1))$ answer different questions.

Numerical Example

Outcome rule: $Y = 120 + 10C - 6T$

Patient A
$C = +1\ (sicker)$

Patient B
$C = -1\ (healthier)$

Observed comparison: $124 - 110 = +14$

But treatment itself lowers Y by 6. Baseline severity created the misleading observed difference.

Three Different Questions

1. Association
What is Y among people we observe with $T = 1$ versus $T = 0$?

$$P(Y \mid T = t)$$

2. Intervention
What would Y be if we forced $T = t$ for the target population?

$$P(Y \mid do(T = t))$$

3. Counterfactual
For this same person, what would Y have been under another treatment?

$$Y(t)$$

SCM gives one framework for all three, but they are not the same question.

Counterfactual: Same person, Different treatment

Suppose one person has: $C = 1.5 \quad U_Y = 2$

Keep these person-specific background factors fixed.

World 1: $T = 1$

$$Y(1) = 120 + 10(1.5) - 6 + 2 = 131$$

World 0: $T = 0$

$$Y(0) = 120 + 10(1.5) + 2 = 137$$

$$\mathrm{Y}(1) - \mathrm{Y}(0) = -6$$

g-Computation does the same thing

1
set T = 1 and predict Y

2
set T = 0 and predict Y

3
average the difference

That procedure is trying to emulate the two intervention worlds described by the SCM.

The estimation model is not the SCM itself, but the intervention logic is the same.

Simulation: Build Data from SCM

$$C \sim \text{Normal}(0,1)$$

$$P(T=1 \mid C) = \text{logistic}(1.2C)$$

$$Y = 120 + 10C - 6T + \varepsilon$$

Interpretation:

- Sicker patients (larger C) are more likely to receive treatment.
- Sicker patients also have higher Y.
- Treatment truly lowers Y by 6 units.

True intervention effect = −6

Simulation: Build Data from SCM

```
set.seed(7900)
  n <- 5000

  C <- rnorm(n)
p <- plogis(1.2*C)
T <- rbinom(n, 1, p)

 eps <- rnorm(n, sd = 5)
Y <- 120 + 10*C - 6*T + eps

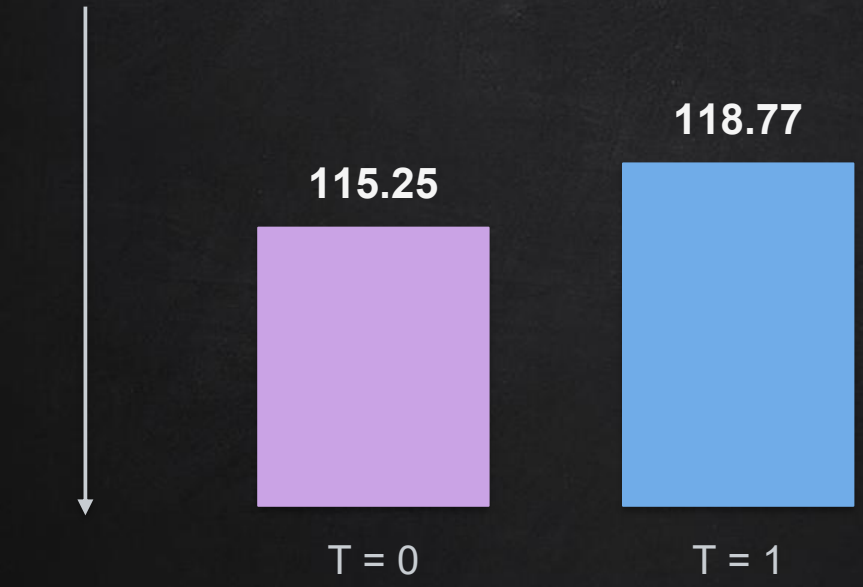
dat <- data.frame(C, T, Y)
```

Before fitting anything, predict:

- Will treated and untreated patients have the same average C?
- Will the crude difference in Y recover -6?
- What should happen after we adjust for C?

Simulation: Build Data from SCM

Observed mean Y



Crude observed difference:

$$118.77 - 115.25 = +3.52$$

But the true treatment effect is −6.

Wrong sign!

One simulated realization; exact sample numbers vary slightly with the random-number generator.

Simulation adjust for Common Cause C

```
  fit_crude <- lm(Y ~ T, data = dat)
fit_adj   <- lm(Y ~ T + C, data = dat)

        coef(fit_crude)["T"]
            # about +3.5

         coef(fit_adj)["T"]
            # about -6.0
```

Crude model
T coefficient ≈ +3.5

Adjusted model
T coefficient ≈ −6.0

The adjusted model recovers the truth here because the simulation satisfies the required assumptions and the outcome rule is correctly represented.

Simulation creates two interventions

```
# keep each person's C and eps fixed
    Y1 <- 120 + 10*C - 6*1 + eps
    Y0 <- 120 + 10*C - 6*0 + eps

     mean(Y1)      # about 114.0
     mean(Y0)      # about 120.0
      mean(Y1 - Y0) # exactly -6
```

do(T = 1)
mean Y ≈ 114.0

do(T = 0)
mean Y ≈ 120.0

$$\text{ATE} = 114 - 120 = -6$$

Real data example: job training and later earnings

Causal question: Does participation in the National Supported Work job-training program increase earnings in 1978?

LaLonde / MatchIt data

614	185	429
people	treated	controls

Treatment T: job-training participation
Outcome Y: 1978 earnings (re78)

What was known before treatment?

Age • education • race • marital status
No high-school degree • earnings in 1974 • earnings in 1975

Call these baseline variables X

These variables may affect both program participation and later earnings.

What does SCM add?

Treated	Control	Crude difference
$6,349	$6,984	−$635

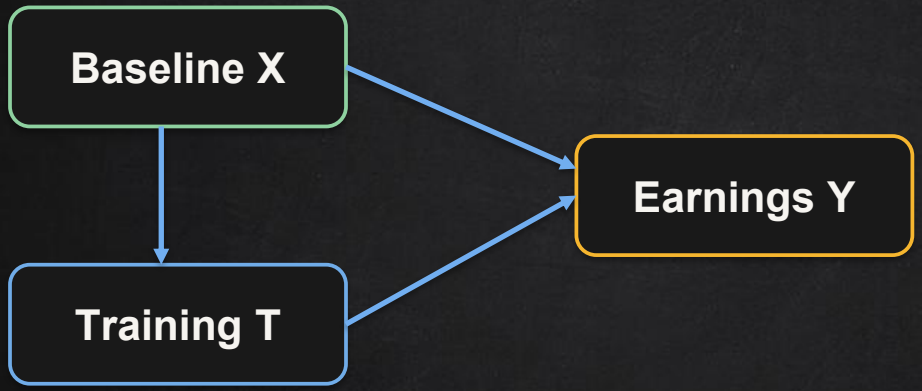
But the groups were already different before treatment

Baseline variable	Treated	Control
Age	25.8	28.0
Married	18.9%	51.3%
Black	84.3%	20.3%
1974 earnings	$2,096	$5,619
1975 earnings	$1,532	$2,466

So −$635 is an observed association, not automatically the causal effect of training.

What do observed data say?

A plausible causal story



What we observed

$$E[Y \mid T = 1] - E[Y \mid T = 0]$$

What we want causally

$$E[Y \mid do(T = 1)] - E[Y \mid do(T = 0)]$$

SCM tells us what “force training” versus “force no training” means. It does not make −$635 causal; assumptions and appropriate adjustment are still needed.

Next

SCM gave us a precise target:

$$E[Y \mid do(T=1)] - E[Y \mid do(T=0)]$$

Next question:

When can we learn that intervention effect from the data we actually observe?

Next: randomized experiments and identification

Bridge

SCM gave us a target

$$E[Y \mid do(T = 1)] - E[Y \mid do(T = 0)]$$

Observed data give us

$$E[Y \mid T = 1] - E[Y \mid T = 0]$$

Identification asks

When are these two comparisons equal?

Identification Question

Causal estimand

$$E[Y(1)] - E[Y(0)]$$

→

Observed distribution

$$E[Y \mid T = 1] - E[Y \mid T = 0]$$

→

Identification

A set of assumptions that lets us rewrite the causal estimand using the observed-data distribution.

Identification is not an estimator. It is the logic that tells us what observable quantity represents the causal target.

Association vs Intervention

Observed association

Who is observed with $T = 1$ versus $T = 0$?

$$P(Y \mid T = t)$$

Intervention distribution

What would happen if the same target population were assigned $T = t$?

$$P(Y \mid do(T = t))$$

Randomization is a design that can make these distributions coincide for the causal question of interest.

When can we replace $do(T = t)$ by $T = t$

$$P(Y \mid do(T{=}t)) = P(Y \mid T{=}t)$$

This equality is NOT true automatically.

One route

Randomize treatment assignment.

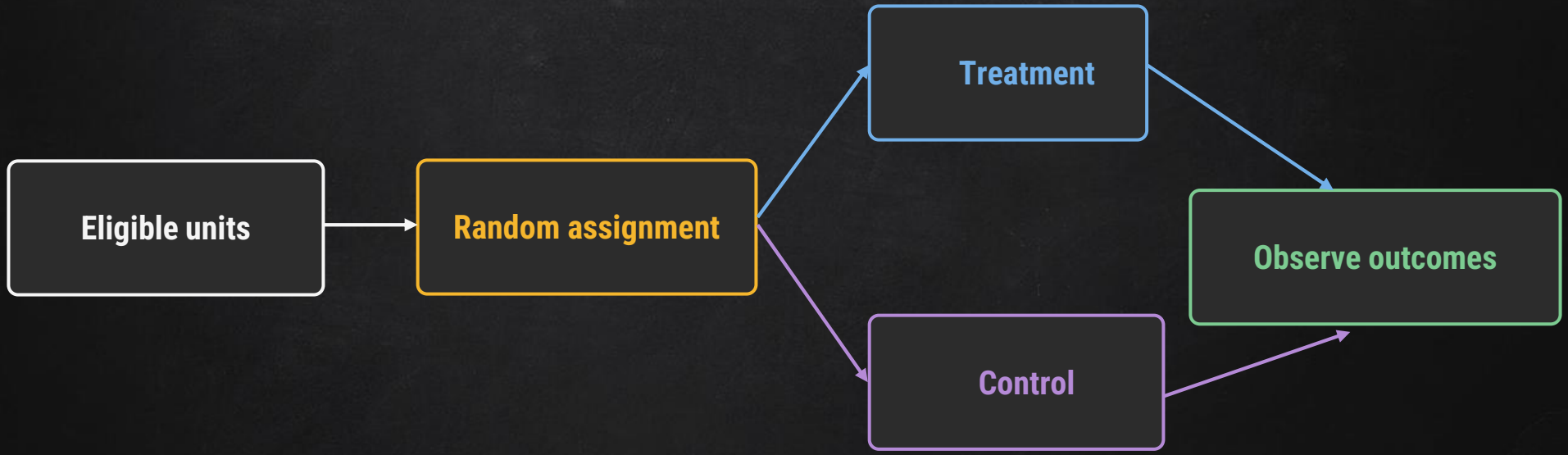
Another route

In observational data, condition on a sufficient adjustment set.

Other routes

IV, front-door, DiD, synthetic control, etc.

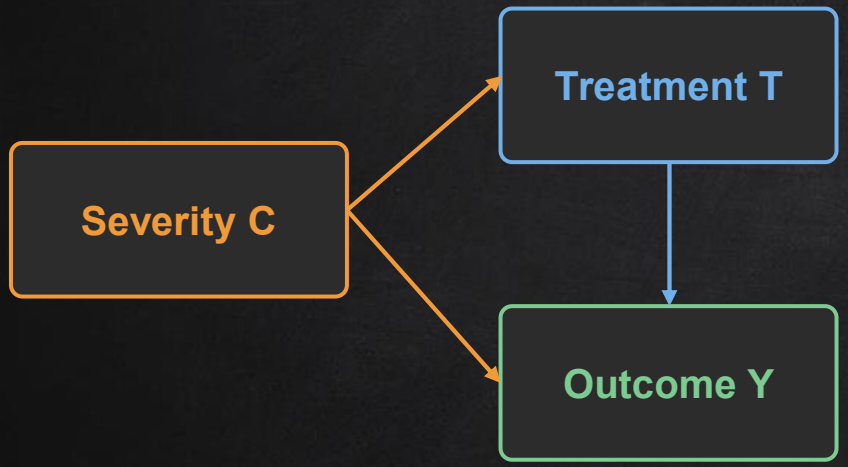
Randomized Experiment: Core Idea



The treatment mechanism is deliberately created by the investigator rather than chosen by prognosis, preference, or policy.

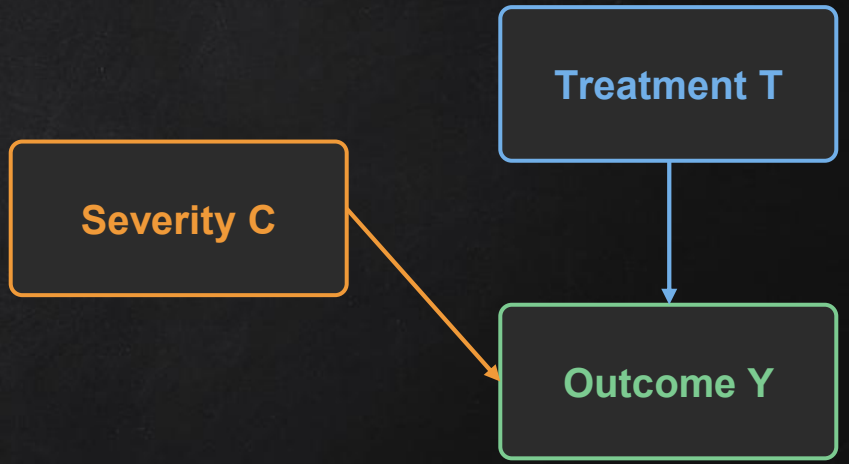
Randomized Experiment: Core Idea

Observational treatment assignment



$T \leftarrow C \rightarrow Y$

Randomized treatment assignment



No systematic C → T arrow

Randomized Experiment in Potential Outcome

$$T \perp \{Y(1), Y(0), X\}$$

Meaning

The assigned treatment does not depend on potential outcomes or baseline covariates.

Consequence

Treated and control units are valid stand-ins for each other's missing counterfactual outcomes, on average over randomization.

This is exchangeability created by design.

ATE Identification in a Randomized Trial

$$ATE = E[Y(1)] - E[Y(0)]$$

↓ under randomization + consistency

$$ATE = E[Y \mid T = 1] - E[Y \mid T = 0]$$

Design result

A causal estimand becomes an observed mean contrast.

Analysis result

Difference in sample means is the natural estimator.

Randomization solves the confounding problem. It does not eliminate sampling uncertainty.

Randomization does not eliminate ALL assumptions

Consistency

Treatment versions must match the intervention being defined.

Positivity

Each unit must have a nonzero chance of assignment to relevant arms.

No interference

One unit's assigned treatment should not alter another unit's outcome for a standard individual-level estimand.

Outcome measurement

Outcomes must be measured comparably enough to support the target contrast.

Randomization primarily creates exchangeability. The rest of the causal story still matters.

Design vs Analysis

Design stage

- Define target population
- Define treatment versions
- Choose randomization scheme
- Pre-specify outcomes and estimands
- Protect against interference / contamination
- Plan sample size and follow-up

Analysis stage

- Estimate arm-specific outcome means
- Compare according to estimand
- Quantify uncertainty
- Adjust for baseline covariates if appropriate
- Handle missing outcomes / noncompliance carefully
- Report effect + interval, not only p-value

In causal inference, strong design often does more work than a complicated analysis.

Randomized Trial Example

Population

Adults with uncontrolled hypertension meeting trial eligibility criteria.

Treatment

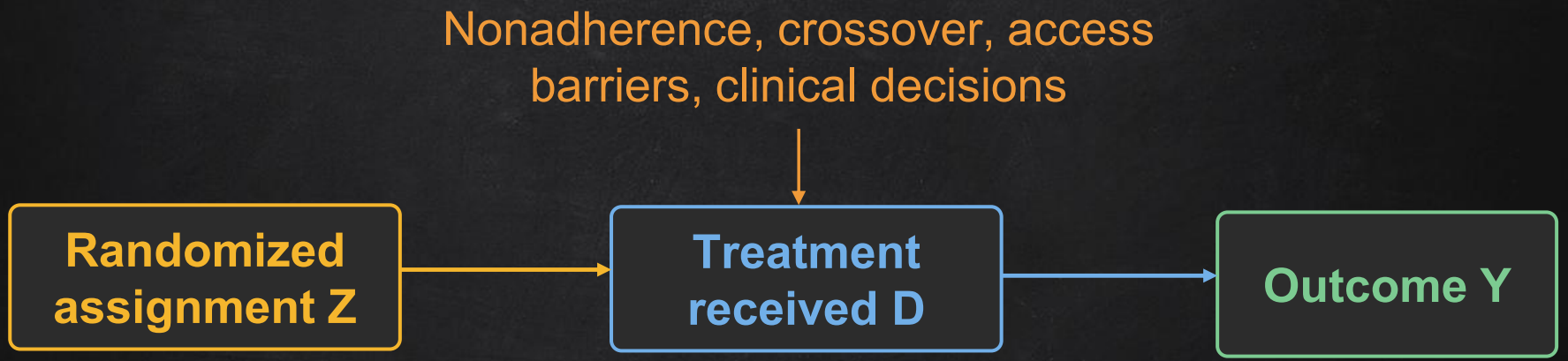
T=1: new drug once daily
T=0: placebo / usual care
Randomize 1:1

Outcome

Systolic blood pressure after 4 weeks.

Target: average treatment effect in trial participants.

Assignment is NOT same as treatment received



Randomization guarantees exchangeability for assignment Z. It does not automatically randomize treatment actually received D.

Complete Randomization

Design

Fix the number assigned to each arm, then randomly choose which units receive each arm.

Example: $n = 200$ → exactly 100 treatment and 100 control.

Why use it?

- Simple
- Exact arm sizes
- Transparent assignment mechanism
- Basis for many randomization-based results

$$\Pr(T_i = 1) = 100/200 = 0.5$$

Bernoulli Randomization

Design

Each unit is randomized independently, e.g.

$T_i \sim \text{Bernoulli}(0.5)$

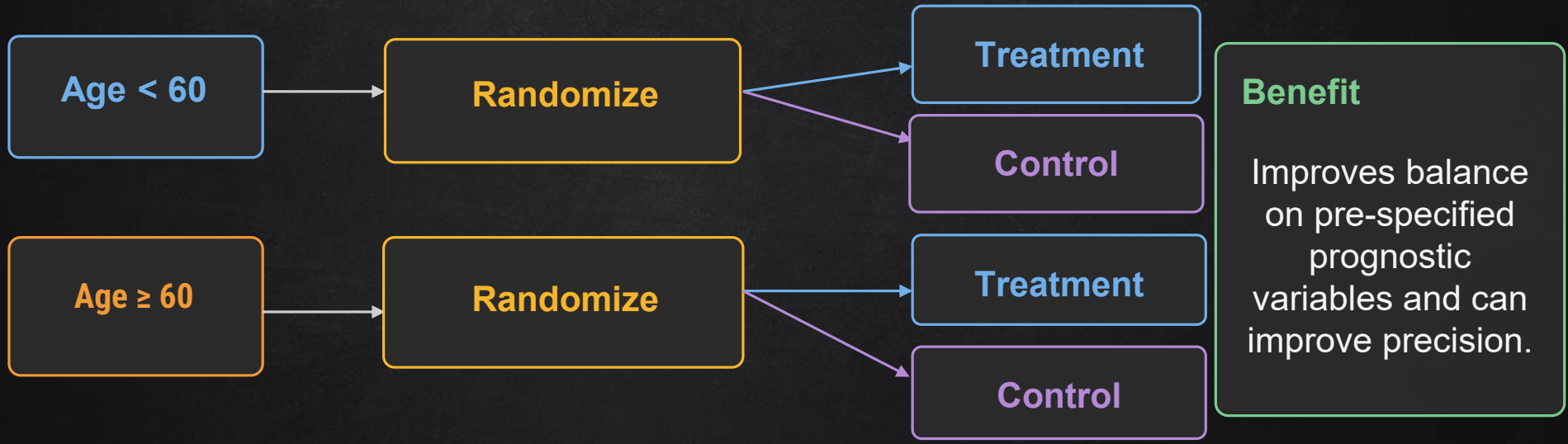
Difference from Complete

The final treatment-arm size is random. You might end with 96 treated and 104 controls, or 107 and 93, etc.

Both are randomized designs; they induce slightly different assignment distributions.

Blocked / Stratified Randomization

Randomize within important baseline groups rather than across everyone at once.



Stratification is a design choice, not a post hoc repair for confounding.

Cluster Randomization

Sometimes we randomize groups rather than individuals.

Clinic 1	Clinic 2	Clinic 3	Clinic 4
Treatment arm	Treatment arm	Control arm	Control arm

Why? To reduce contamination, simplify implementation, or because the intervention operates at group level.

Analysis must account for within-cluster correlation; the effective sample size is smaller than the raw number of individuals suggests.

Matched Pair Randomization

Form pairs of similar units first, then randomize one unit in each pair to treatment.



Useful when

A strong prognostic variable is known and matching can make within-pair comparisons much more precise.

What randomization guarantees

Known assignment probabilities

We know how treatment was generated.

No systematic confounding by baseline variables

Assignment is independent of baseline prognosis by design.

Unbiased treatment contrast over repeated randomizations

The design centers the difference-in-means estimator at the causal effect.

These are properties of the assignment mechanism, not promises about one realized dataset.

What randomization does not guarantee

Exact covariate balance

Chance imbalance can occur.

No missing outcomes

People can still drop out or miss visits.

Perfect adherence

Assigned treatment may not be taken.

No interference

People or clusters can affect each other.

Generalizability

Trial volunteers may differ from target population.

Large sample precision

A small RCT can be unbiased but very noisy.

A Baseline Balance Table

Variable	Control	Treatment	Difference
Age, years	58.2	57.6	+0.6
Baseline SBP	128.4	131.6	−3.2
Female	52%	49%	+3 pp
Diabetes	21%	18%	+3 pp

Question

Should we “fix” every visible baseline difference?

Use pre-specified covariate adjustment for precision, not data-driven fishing for variables with small baseline p-values.

Exchangeability vs Covariate Balance

Exchangeability

A statement about the assignment mechanism and potential outcomes.

In a correctly randomized experiment, it holds by design before we look at the data.

Observed balance

A descriptive feature of one realized randomization.

It can look imperfect even when exchangeability is valid.

SATE vs PATE

Sample Average Treatment Effect (SATE)

Average causal effect for the units actually enrolled in this experiment.

Finite-sample target.

Population Average Treatment Effect (PATE)

Average causal effect for a broader target population from which trial participants are viewed as a sample.

Requires a sampling / transport story.

Randomization identifies the treatment contrast inside the trial. Generalization beyond the trial is a separate problem.

From Identification to Estimation

Causal target

$$ATE = E[Y(1)] - E[Y(0)]$$

Randomization + consistency

$$= E[Y \mid T = 1] - E[Y \mid T = 0]$$

Estimation

$$\approx \bar{Y}_1 - \bar{Y}_0$$

Identification told us WHICH observed-data quantity equals the causal estimand.
Estimation tells us how to learn that quantity from a finite sample.

Next, we finish the randomized-trial story: point estimate → uncertainty → inference.

The Difference-in-Means Estimator

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$$

$\bar{Y}_1$

Average observed outcome among participants assigned to treatment.

$\bar{Y}_0$

Average observed outcome among participants assigned to control.

$\hat{\tau}$

Estimated average treatment effect in the trial sample.

For a continuous outcome and a simple two-arm randomized trial, this is the natural estimator.

Simple analysis is enough because the design already did the confounding control.

Worked Example: Continuous Outcome

Outcome: systolic blood pressure after 4 weeks

Treatment arm

$n_1 = 100$

Mean SBP = 126.2 mmHg

Control arm

$n_0 = 100$

Mean SBP = 131.5 mmHg

$$\hat{\tau} = 126.2 - 131.5 = -5.3\ mmHg$$

Interpretation: assignment to the new treatment lowered mean 4-week SBP by 5.3 mmHg in this trial sample.

Always report the direction, magnitude, and units of the effect.

Why does Difference in Means Work?

Treatment group

Randomization makes this group a valid sample of the potential outcomes under treatment.

Control group

Randomization makes this group a valid sample of the potential outcomes under control.

Difference

Subtracting the two arm means estimates the average causal contrast.

The estimator is simple because exchangeability came from the randomization mechanism.

Standard Error of the Difference in Means

$$SE(\hat{\tau}) = \sqrt{(s_1^2/n_1 + s_0^2/n_0)}$$

s_1 and s_0

Sample standard deviations within the treatment and control arms.

n_1 and n_0

Numbers of participants observed in the two treatment arms.

$SE(\hat{\tau})$

Typical sampling variability of the estimated difference in means.

More outcome variability increases SE; larger sample sizes decrease SE.

Worked Example: 95% Confidence Interval

Suppose $\hat{\tau} = -5.3$ mmHg and $SE(\hat{\tau}) = 1.8\ mm$Hg.

$$\mathbf{95\%\ CI} \approx \hat{\tau} \pm \mathbf{1.96} \times \boldsymbol{SE}(\hat{\tau})$$

$$= -5.3 \pm 1.96(1.8)$$

$$= \mathbf{(-8.8, -1.8)\ mmHg}$$

The data are compatible with a moderate reduction in SBP; the interval excludes 0 in this example.

Report the effect estimate together with its interval, not only a p-value.

What Does a 95% Confidence Interval Mean?

Correct repeated-sampling idea

If we repeated the trial many times using the same procedure, about 95% of similarly constructed intervals would contain the true target effect.

Common incorrect statement

“There is a 95% probability that the true effect lies in this particular interval.”

In the frequentist framework, the fixed effect is not assigned a probability after the interval is observed.

Confidence describes the long-run performance of the interval procedure.

Hypothesis Test and p-value

$$H_0: \tau = 0 \quad \text{versus} \quad H_1: \tau \neq 0$$

A p-value asks: if the null hypothesis were true, how unusual would an estimate at least this extreme be under the assumed analysis?

Small p-value

Data are difficult to reconcile with H_0 under the analysis assumptions.

Large p-value

Does NOT prove “no effect”; the trial may simply be imprecise.

Effect size + confidence interval usually communicates more than the p-value alone.

Randomization Inference

Inference can come directly from the randomization mechanism.

1

Assume a sharp null: treatment changes no participant's outcome.

2

Re-randomize treatment labels using the same design used in the trial.

3

Recompute the difference in means for each random assignment.

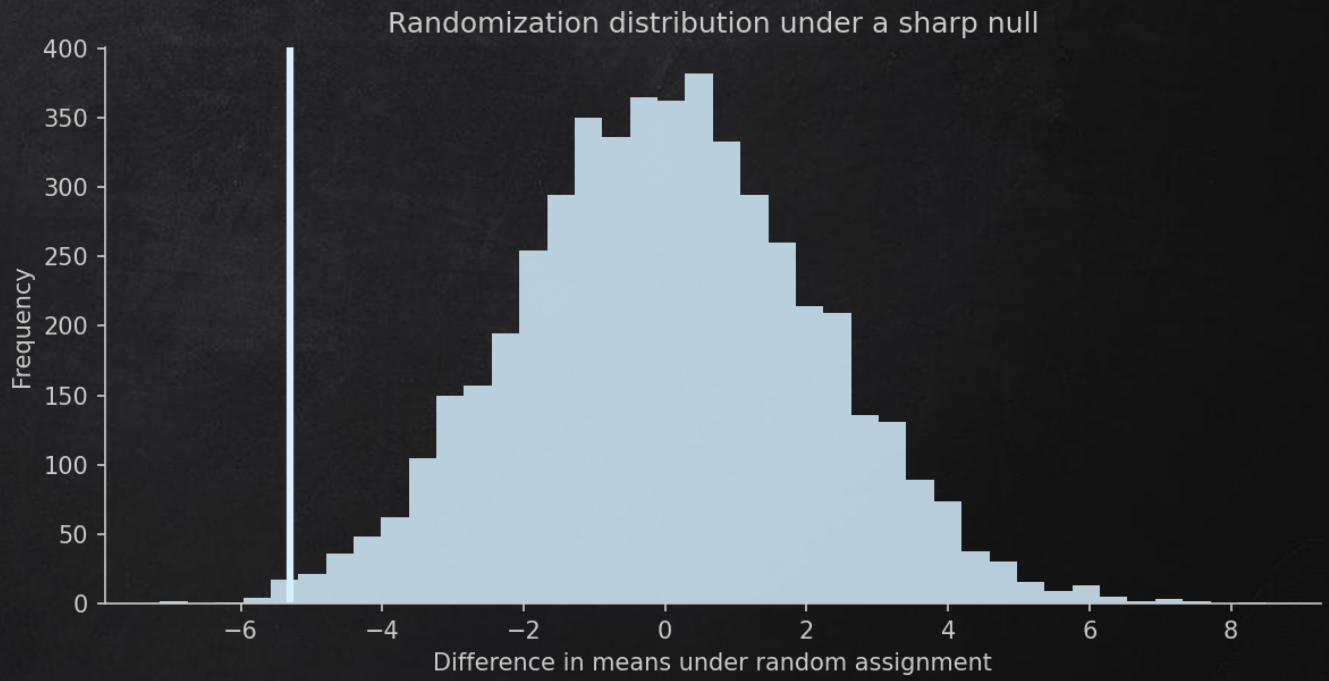
4

Compare the observed difference with the randomization distribution.

This test uses the experiment's actual assignment mechanism rather than a model for outcomes.

Where Is the Observed Effect?

If −5.3 lies far in the tail of the null distribution, such a difference is unusual under no treatment effect.



Randomization inference turns “what else could random assignment have produced?” into a formal test.

Where Is the Observed Effect?

$$Y_i = \beta_0 + \beta_1 T_i + \varepsilon_i$$

β_0

Mean outcome in the control arm when T=0.

β_1

Difference between treatment and control means: $\bar{Y}_1 - \bar{Y}_0$.

$$\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$$

With only a binary treatment indicator, OLS and the raw difference in means are the same estimator.

Where Is the Observed Effect?

```
# Difference in means
mean(dat$Y[dat$T == 1]) -
  mean(dat$Y[dat$T == 0])

# Same estimate from regression
fit <- lm(Y ~ T, data = dat)
coef(fit)["T"]
```

Same causal estimate

In a simple randomized two-arm trial, both approaches estimate the treatment-control mean difference.

Regression becomes useful when we add covariates, interactions, or design features.

R syntax changes; the estimand should not.

Why adjust for baseline covariates in RCT?

Not to “remove confounding” created by baseline variables.

Reason 1

Improve precision when a baseline variable strongly predicts the outcome.

Reason 2

Account for pre-specified stratification / design variables appropriately.

Reason 3

Protect against chance imbalance without data-driven covariate hunting.

Randomization gives validity; sensible baseline adjustment can give efficiency.

Covariate-Adjusted Analysis: ANCOVA

$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 X_i + \varepsilon_i$$

Example: X = baseline systolic blood pressure.

Treatment coefficient β_1

Adjusted treatment-control difference, holding the pre-treatment covariate fixed.

Key restriction

Use baseline/pre-treatment covariates. Do not “adjust” for post-treatment mediators when targeting the total effect.

This connects directly to the DAG logic from earlier lectures.

Adjusted vs Unadjusted

Illustrative results from the same randomized trial

Analysis	Estimate	SE	95% CI
Unadjusted	−5.3	2.1	(−9.4, −1.2)
Adjusted for baseline SBP	−5.1	1.6	(−8.2, −2.0)

The point estimate barely changes, but the adjusted analysis is more precise because baseline SBP explains outcome variation.

Do not choose adjustment variables because they “look imbalanced”; pre-specify important prognostic covariates.

Binary Outcomes: Risk in Each Arm

Suppose the outcome is hospitalization by 30 days.

Treatment

12 hospitalizations / 100
Risk = 0.12 = 12%

Control

20 hospitalizations / 100
Risk = 0.20 = 20%

The two most intuitive causal contrasts are the risk difference and risk ratio.

For binary outcomes, start by estimating the event risk in each randomized arm.

Risk Difference

$$\mathrm{RD} = \mathrm{Risk}_1 - \mathrm{Risk}_0$$

$$= 0.12 - 0.20 = -0.08$$

Interpretation: treatment reduced the 30-day hospitalization risk by 8 percentage points in this trial.

Absolute effect

Risk difference is on the probability scale and often easiest to communicate clinically.

−0.08 means 8 fewer events per 100 treated participants, on average.

Risk Ratio

$$RR = Risk_1 / Risk_0$$

$$= 0.12 / 0.20 = 0.60$$

Interpretation: the hospitalization risk under treatment was 60% of the risk under control.

Relative effect

Equivalently, this example corresponds to a 40% relative risk reduction: 1 − 0.60 = 0.40.

Absolute and relative effects answer different communication questions; report the one tied to your estimand.

Risk Ratio

```
# Risks by arm
risk1 <- mean(dat$Y[dat$T == 1])
risk0 <- mean(dat$Y[dat$T == 0])

# Risk difference
RD <- risk1 - risk0

# Risk ratio
RR <- risk1 / risk0
```

Interpret first, model second

When Y is coded 0/1, the sample mean in each arm is the estimated risk.

So, RD and RR can be computed directly before introducing logistic regression or odds ratios.

Keep the estimand visible: risk difference and risk ratio are not the same causal measure.

Summary



$$E[Y(1)] - E[Y(0)] \rightarrow E[Y|T=1] - E[Y|T=0] \rightarrow \bar{Y}_1 - \bar{Y}_0$$

Next class: estimation when treatment was NOT randomized.

Randomized trials are the clean benchmark. Observational estimation must recreate the missing comparability through assumptions + adjustment.